$A=\begin{bmatrix}3&0\\0&1\end{bmatrix}$ $U=I_2$ $\Sigma=\begin{bmatrix}3&0\\0&1\end{bmatrix}$ $V=I_2$

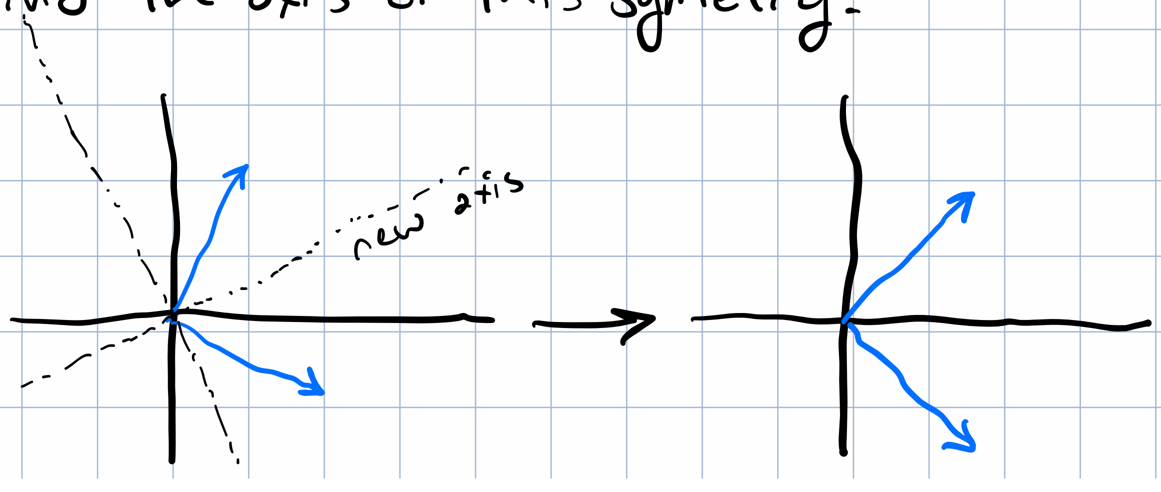
$A=U\Sigma V^T$

1) V^T rotates/reflects the vector acted upon into a coordinate system where the Σ transformation is just scaling, and then rotating back in to the origional coordinate system via U.

2) If $\Sigma=\begin{bmatrix}5&0\\0&2\end{bmatrix}$ then the principal axes are scaled by 5 (for x, or the first axis) and 2 (for y) respectively

3) There is a reduction of dimension, as the axis will be collapsed down to zero.

4) The eigenvectors of A^TA are always orthogonal because A^TA is symetric. The eigenvectors find the axis of this symetry.



5 If the data matrix has a $\Sigma = \begin{bmatrix} 100 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ we should

keep the first column if we want a rank-1 approximation as it accounts for about 80% of the corrolations in the data ($^{100}/_{121}$). The second column accounts for about 19% ($^{20}/_{121}$), and the third for less than 1%, and therefore those columns should be discarded.

A rank-2 approximation would keep the first two, accounting for about 99% of the corrolations.